Player-1: Learning to Play from Synthetic Experience

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Paper version: arXiv 2610.20584v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Demonstrations cover only a limited portion of the states an agent may encounter.*

Can **synthetic experience** push game control beyond SFT, and transfer to games outside the RL suite? The paper tests on three external benchmarks: MineDojo, ViZDoom and GameWorld, none part of the RL suite.

Motivation: three challenges

1. **Mixed demos.** Input conventions differ by source.
2. **Rewards.** Hard to write for commercial games.
3. **One policy.** Dynamics and reward scales differ.

Evaluation setup

MineDojo-300 · 300 tasks | ViZDoom-10 · 10 maps | GameWorld · 170 tasks | Qwen3.5-9B / 35B-A3B | 200 RL steps | 256 episodes per step | 8 / 50 / 3 trials

- **All online** rollouts; metrics differ by benchmark.
- **Outside the RL suite:** all three; Minecraft is in the SFT demos.

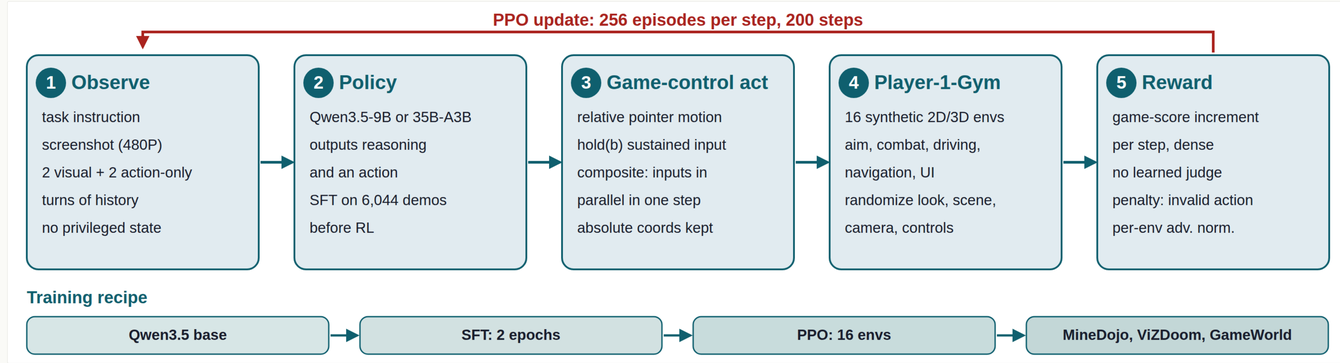
Our read (CUA Radar)

75	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; games outside the RL suite; 95% CI (MineDojo only); SFT-stage baselines; component ablation; budget and cost partly.
- **Flags:** AP2 (synthetic training environments).
- Pattern K13. Authors' numbers, not rerun.

Method: shared action space, SFT, then PPO in 16 synthetic gyms

- **Sees / acts:** screenshot, task text and history in; reasoning plus keyboard and mouse actions out.
- **New:** relative pointer motion, sustained holds, composite (concurrent) actions.
- **Signal:** game-score increments from privileged state, no learned judge.



Our redraw (not the paper's figure): the observe, act, reward loop and the training recipe.

$$\hat{A}^{e}_{i,j} = \frac{A_{i,j} - \mu_e}{\sqrt{\sigma_e^2 + \epsilon}}$$

Per-environment advantage normalisation (Eq. 1)

Key results: MineDojo-300 success rate (%)

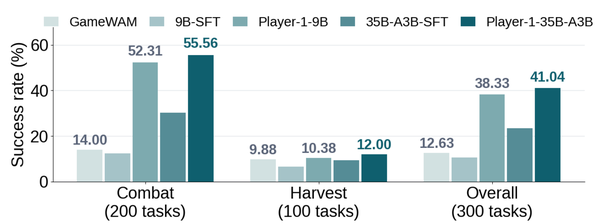
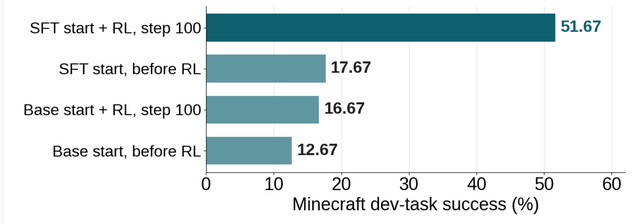


Table 2, redrawn: 8 trials per task; the paper reports 95% CIs for the overall column.

Player-1-9B 38.33% vs 12.63% for GameWAM (strongest external)

Ablation: SFT start vs base start, 9B



Section 5.1 / Fig. 5c, redrawn: Minecraft dev-task success (%) at the start and after 100 RL steps.

SO WHAT → *SFT start matters (51.67 vs 16.67 at step 100); the base start moves from 12.67 to 16.67. The paper's Fig. 7 also shows success rising as RL uses more environments.*

Headline Numbers

38.33%
MineDojo-300 success, 9B
vs 10.54% after SFT; GameWAM 12.63%

- **41.04%** 35B-A3B, MineDojo
- **0.6922** ViZDoom-10 (9B)
- **34.45%** GameWorld progress

Controls (all online)

CONTROL	PLAYER-1	COMPARED WITH
ViZDoom-10 return, 9B	0.6922	SFT 0.4361
ViZDoom-10, 35B-A3B	0.7236	GameWAM 0.3938
GameWorld progress %, 9B	34.45	SFT 31.43
ViZDoom mobile combat, 9B	0.3033	SFT 0.5293

GameWorld gains are small

Takeaway

SFT plus PPO in 16 synthetic gyms lifts MineDojo-300 success to 38.33% (9B) and 41.04% (35B-A3B); the other benchmarks gain less.

6,044 demos | 16 synthetic envs | composite use 64.7% → 93.2%

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